

51—(MathAA/10_HL_Winter_2022_Q4)—Integration

[Maximum mark: 5]

The derivative of the function f is given by $f'(x) = \frac{6x}{x^2 + 1}$.

The graph of $y = f(x)$ passes through the point $(1, 5)$. Find an expression for $f(x)$.



1-(MAT/11_HL_Summer_2009_Q13)-ComplexNumbers,DifferentialEquations

Part A [Maximum mark: 9]

If z is a non-zero complex number, we define $L(z)$ by the equation

$$L(z) = \ln|z| + i \arg(z), \quad 0 \leq \arg(z) < 2\pi.$$

- (a) Show that when z is a positive real number, $L(z) = \ln z$. [2 marks]
- (b) Use the equation to calculate
- (i) $L(-1)$;
- (ii) $L(1-i)$;
- (iii) $L(-1+i)$. [5 marks]
- (c) Hence show that the property $L(z_1 z_2) = L(z_1) + L(z_2)$ does not hold for all values of z_1 and z_2 . [2 marks]

Part B [Maximum mark: 14]

Let f be a function with domain $\mathbb{R}$ that satisfies the conditions,

$$f(x+y) = f(x)f(y), \text{ for all } x \text{ and } y \text{ and } f(0) \neq 0.$$

- (a) Show that $f(0) = 1$. [3 marks]
- (b) Prove that $f(x) \neq 0$, for all $x \in \mathbb{R}$. [3 marks]
- (c) Assuming that $f'(x)$ exists for all $x \in \mathbb{R}$, use the definition of derivative to show that $f(x)$ satisfies the differential equation $f'(x) = k f(x)$, where $k = f'(0)$. [4 marks]
- (d) Solve the differential equation to find an expression for $f(x)$. [4 marks]



2-(MAT/10_HL_Winter_2009_Q8)-DifferentialEquations

A certain population can be modelled by the differential equation $\frac{dy}{dt} = k y \cos kt$, where y is the population at time t hours and k is a positive constant.

- (a) Given that $y = y_0$ when $t = 0$, express y in terms of k , t and y_0 . [5 marks]
- (b) Find the ratio of the minimum size of the population to the maximum size of the population. [2 marks]

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3-(MAT/12_HL_Summer_2010_Q14)-DifferentialEquations,Trigonometry

[Maximum mark: 15]

Throughout this question x satisfies $0 \leq x < \frac{\pi}{2}$.

- (a) Solve the differential equation $\sec^2 x \frac{dy}{dx} = -y^2$, where $y = 1$ when $x = 0$.

Give your answer in the form $y = f(x)$.

[7 marks]

- (b) (i) Prove that $1 \leq \sec x \leq 1 + \tan x$.

- (ii) Deduce that $\frac{\pi}{4} \leq \int_0^{\frac{\pi}{4}} \sec x \, dx \leq \frac{\pi}{4} + \frac{1}{2} \ln 2$.

[8 marks]



4-(MAT/10_HL_Winter_2011_Q13)-Differential Equations

The curve C with equation $y = f(x)$ satisfies the differential equation

$$\frac{dy}{dx} = \frac{y}{\ln y} (x + 2), \quad y > 1,$$

and $y = e$ when $x = 2$.

- (a) Find the equation of the tangent to C at the point $(2, e)$. [3 marks]
- (b) Find $f(x)$. [11 marks]
- (c) Determine the largest possible domain of f . [6 marks]
- (d) Show that the equation $f(x) = f'(x)$ has no solution. [4 marks]



5-(MathAA/10_HL_Winter_2021_Q8)-DifferentialEquations

[Maximum mark: 7]

Solve the differential equation $\frac{dy}{dx} = \frac{\ln 2x}{x^2} - \frac{2y}{x}$, $x > 0$, given that $y = 4$ at $x = \frac{1}{2}$.

Give your answer in the form $y = f(x)$.

